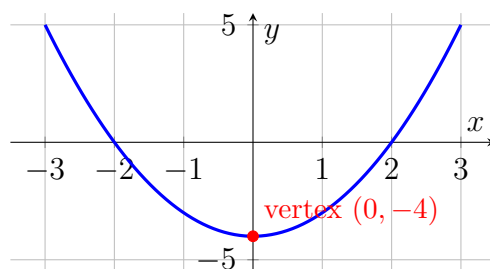


Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

Quadratic Functions: Vertex, Max & Min

Example. For $y = x^2 - 4$ the vertex is $(0, -4)$, a **minimum**. The graph *decreases* for $x < 0$ and *increases* for $x > 0$.



Directions: for each function state max or min, give the vertex, and the increasing / decreasing intervals.

1. $y = x^2 + 2$

3. $y = (x - 3)^2$

2. $y = -x^2 + 5$

4. $y = -(x + 1)^2 - 2$

Challenge (same sheet):

5. $y = (x - 2)^2 - 9$ — give the vertex, both intervals, *and* the two x -intercepts.

Answer Key — fully worked

Procedure: opens up $(+x^2) \Rightarrow$ minimum; opens down $(-x^2) \Rightarrow$ maximum. From $y = a(x - h)^2 + k$ the vertex is (h, k) ; the graph turns there.

1. $y = x^2 + 2$: opens **up** $\Rightarrow$ **min**; vertex $(0, 2)$; decreasing $x < 0$, increasing $x > 0$.
2. $y = -x^2 + 5$: opens **down** $\Rightarrow$ **max**; vertex $(0, 5)$; increasing $x < 0$, decreasing $x > 0$.
3. $y = (x - 3)^2$: vertex $(3, 0)$, opens up $\Rightarrow$ **min**; decreasing $x < 3$, increasing $x > 3$.
4. $y = -(x + 1)^2 - 2$: vertex $(-1, -2)$, opens down $\Rightarrow$ **max**; increasing $x < -1$, decreasing $x > -1$.
5. $y = (x - 2)^2 - 9$: vertex $(2, -9)$ **min**; dec $x < 2$, inc $x > 2$; intercepts: $(x - 2)^2 = 9 \Rightarrow x - 2 = \pm 3 \Rightarrow x = -1, 5$.